

A Fixed Claim on a Shrinking Supply: AI Data Centres, Water Stress and Lock-In

Evidence from global freshwater use and the Nile basin

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Abstract

Public discussion increasingly presents artificial intelligence as a technology that takes water from people. This paper tests that claim against the scale of human freshwater use and examines where AI infrastructure is being built. Data centres withdrew about 5.5 km³ of fresh water in 2023, roughly 0.14% of human withdrawals; livestock production consumes about 270 times as much surface and groundwater, and leaking supply networks lose about 23 times what data centres withdraw. At global scale, AI is not a cause of freshwater scarcity. The risk is local and lies in siting: about half of the world's data-centre hubs, and two-thirds of new US data centres since 2022, are in high water-stress areas; local utility burdens at US sites range from 0.2% to over 100% of host capacity; and facilities designed to run for 15 to 20 years (industry estimates) are being placed in basins with no reserve and, in Egypt, a single external source of supply. Read as a system, this is a failure of governance and pricing. The paper proposes three tracks: non-potable cooling water and low-water electricity in water-stressed basins, with renewable-powered desalination of all on-site water costing less than 0.2% of the sector's projected electricity; siting assessments over each facility's operating life, with complete water reporting; and AI applied to reducing losses in local water networks, additional to work utilities already plan.

Keywords: artificial intelligence; data centres; freshwater; water stress; infrastructure siting; lock-in; desalination; commons; Nile; systems resilience

Key findings and recommendations

Finding	Evidence
AI is a small user of water at global scale	Data centres withdraw ~0.14% of human withdrawals; agriculture takes 72%
Other losses dwarf it	Network losses are ~23× data-centre withdrawals; livestock consumes ~270× as much surface and groundwater
The risk is siting, not volume	Half of global hubs and two-thirds of new US sites are in high-stress areas; US local burdens range up to 134% of utility capacity
The exposure is greatest where supply has no alternative	Egypt draws over 98% of its renewable water from abroad while recruiting data-centre investment
The fix is affordable	Desalinating all on-site data-centre water would take under 0.2% of the sector's projected electricity

Track	Core recommendation
Track A	Non-potable cooling water (recycled, seawater or renewable-powered desalination with brine limits) and low-water electricity in stressed basins
Track B	Permits assess water supply over the facility's life; operators report on-site and indirect water by source
Track C	Operators fund additional AI-assisted loss reduction in the same basin, at least equal to the water they draw from it

1. Introduction

Since a 2024 analysis by the *Washington Post* with researchers at the University of California, Riverside estimated that GPT-4 required about 519 millilitres of water, roughly a bottle, to write a 100-word email (Verma and Tan 2024), the image of AI "drinking" water has circulated widely. Per-query estimates, however, differ by orders of magnitude. Google's in-production measurement put the median Gemini text prompt at 0.26 millilitres of on-site water consumption, against earlier published estimates of 45 to 50 millilitres (Elsworth et al. 2025). Much of the difference lies in what is counted: the higher figures include water used to generate electricity, the lower ones only water consumed at the data centre (Section 2.3). None of these figures, on its own, shows whether AI is a cause of freshwater scarcity. That question can only be answered by comparison with the scale of total human water use, which public discussion rarely makes. AI runs in data centres that withdraw and evaporate fresh water, directly for cooling and indirectly through the power plants that supply their electricity. That use is real. It is not, however, what drives global freshwater scarcity.

The distinction matters because the wrong diagnosis leads to the wrong response. If AI is framed as the problem, the answer becomes opposition to AI itself, a technology already embedded in economies and public services. If the problem is framed correctly, the answer is to change how and where AI infrastructure is built, and to treat its water demand as one part of a larger pattern of human water management that is already failing in many regions.

The infrastructure being built in this decade will not be temporary. Power plants, reservoirs, pipelines and data centres sited now will shape how water is used for decades, and the choices embedded in them will pass to people who had no part in making them. Whether that inheritance is a margin of safety or a set of constraints is being decided now.

Recent assessments have quantified AI's resource footprint in detail. The United Nations University Institute for Water, Environment and Health (UNU-INWEH) estimated the carbon, water and land footprints of AI's electricity use across the world's major data-centre hubs (Aczel et al. 2026), and US analyses have mapped data-centre water use against watershed stress (Siddik et al. 2021) and argued that the problem is technically soluble (Gaster 2026). This paper adds four things: a comparison of AI's water demand against the major categories of human freshwater use, including food production and network losses, together with evidence on local utility burden; an analysis of siting and infrastructure lock-in as the mechanism of risk, read through systems theory; a feasibility calculation for supplying data centres' on-site water entirely from renewable-powered seawater desalination; and a set of three prioritised policy tracks, applied to Egypt as a case where water dependence and new data-centre investment now coincide.

Section 2 sets out the methods. Section 3 presents the results, from the global freshwater baseline to the Egyptian case. Section 4 interprets them as a system and examines the institutional conditions under which competition for water escalates. Section 5 sets out the recommendations, Section 6 the limitations, and Section 7 concludes.

2. Methods

2.1 Approach

This is a structured narrative synthesis of published quantitative data, combined with a systems-theoretical analysis. It is not a systematic review. No primary measurements were collected. All figures are drawn from the sources cited, and every value derived by the author is shown with its inputs in Section 2.4.

2.2 Source selection

Sources were selected in order of preference: peer-reviewed studies; reports of United Nations and other intergovernmental bodies (UNU-INWEH, UNESCO/UN-Water, FAO, IEA); government statements and official data; industry and research-organisation reports; and journalism. Lower-tier sources are used only where no higher-tier source was available, mainly for recent events and for data-centre siting patterns, and are identified as such in the text. Bibliographic details were checked against publisher and database records.

Sources were identified in September 2026 through keyword searches of web search engines, publisher platforms and preprint servers, and through the reference lists of key studies. Search terms combined "data centre", "data center" or "artificial intelligence" with "water", "withdrawal", "consumption", "cooling" and "water stress", and combined "Egypt", "Nile" or "GERD" with "water balance", "desalination" and "data centre". Studies were included where they reported quantitative estimates with a stated method or source. Where estimates conflicted, the more recent peer-reviewed estimate was preferred and the alternatives are reported alongside it.

2.3 Measures

Water statistics use three different measures. Confusing them is the most common error in public debate on this topic, and this paper states which measure every figure uses (Table 1).

Measure	Definition	Example
Withdrawal	Water taken from a river, lake or aquifer. Part of it returns to the source.	A cooling system drawing from a municipal supply
Consumption	The part of a withdrawal that does not return, mostly lost to evaporation.	Water evaporated in a cooling tower
Footprint	Consumption plus indirect water used elsewhere in the supply chain, such as for power generation or as rain on crops.	Water evaporated at the power plant supplying a data centre

Table 1. Water measures used in this paper. 1 km³ = 1 billion m³ = 1 trillion litres.

2.4 Derived values

Values derived by the author for this paper are listed with their inputs in Table 2.

Derived value	Calculation	Result
Withdrawals as share of accessible runoff	$4,000 \div 12,500 \text{ km}^3$	~32%
Per-person daily withdrawal	$4,000 \text{ km}^3 \div 8.0 \text{ billion} \div 365$	~1,370 L
Livestock blue water	6.2% of $2,422 \text{ km}^3$	~150 km^3/yr
Livestock blue water vs data-centre consumption, 2023	$150 \div 0.56 \text{ km}^3$	~270×
Data-centre share of human withdrawals	$5.5 \div 4,000 \text{ km}^3$	~0.14%
Network losses vs data-centre withdrawals	$126 \div 5.5 \text{ km}^3$	~23×
On-site data-centre consumption, 2030	25% of 1.2 km^3 (on-site share of IEA 2023 estimate)	~0.30 km^3
On-site data-centre withdrawal, 2030	$0.30 \text{ km}^3 \div 0.8$ (about 80% of drawn water evaporates)	~0.38 km^3
Energy to desalinate on-site withdrawal, 2030	$0.38 \text{ billion m}^3 \times 3.5 \text{ kWh/m}^3$	~1.3 TWh/yr
Desalination energy as share of sector electricity, 2030	$1.3 \div 945 \text{ TWh}$	~0.14%
Annual water cost of desalinated on-site supply	$0.38 \text{ billion m}^3 \times \text{US\$}0.31\text{--}1.50/\text{m}^3$	~US\$0.12–0.56 billion
Additional brine	$0.38 \text{ km}^3 \times 1.5$	~0.56 km^3/yr
Cooling water, evaporatively cooled 1 GW campus	IEA US average (2 million L/day per 100 MW) $\times 10$	~20,000 m^3/day

Table 2. Derived values and their inputs. The on-site share (140 of 560 billion litres in 2023) is from the IEA estimate as reported by de Vries-Gao (2025); the evaporated fraction is from Ren, as reported by Nicoletti et al. (2025); sector electricity for 2030 is the IEA (2025) projection. The 3.5 kWh/m³ figure is the upper end of the current typical range and is used as a conservative assumption. Population is 8.0 billion (UN World Population Prospects 2022).

2.5 Sensitivity

The derived results depend on assumptions that could reasonably be set differently. Table 3 shows each result under alternative values. None of the paper's conclusions changes within these ranges.

Result	Base	Range	Inputs varied
Energy to desalinate on-site withdrawal, 2030	1.3 TWh/yr	0.7–1.8 TWh/yr	1.79–3.5 kWh/m ³ ; on-site share 25–35%
Share of 2030 sector electricity	0.14%	0.07–0.19%	As above
Annual water cost, 2030	US\$0.12–0.56 billion	US\$0.12–0.79 billion	US\$0.31–1.50/m ³ ; on-site share 25–35%
Livestock blue water vs data-centre consumption	~270×	~125–270×	Data-centre consumption 0.56 km^3 (2023) or 1.2 km^3 (2030 projection)
Data-centre share of human withdrawals	0.14%	0.14–0.23%	IEA 2023 withdrawal or UNU-INWEH 2030 footprint
Per-person daily withdrawal	~1,370 L	~1,370–1,390 L	Population 8.0 or 7.9 billion

Table 3. Sensitivity of derived results. The 1.79 kWh/m³ value is the 2025 record for seawater reverse osmosis (Pumps & Systems 2026); the 35% on-site share is an upper-bound test, not an estimate.

2.6 Systems framework

The interpretation in Section 4 draws on three established frameworks: the management of shared resources (Hardin 1968; Ostrom 1990), infrastructure lock-in (Unruh 2000) and the institutional-capacity model of transboundary water conflict (Wolf, Yoffe and Giordano 2003).

3. Results

3.1 How much fresh water exists

About 97.5% of Earth's water is salt water. The remaining 2.5%, around 35 million km³, is fresh, and most of it is locked in ice or deep underground. The portion humans can practically use, in lakes, rivers, soil moisture and shallow groundwater, is about 200,000 km³, less than 1% of all fresh water (UNEP 2002; Figure 1).

All water on Earth (about 1,386 million km³)



Fresh water only (about 35 million km³)



Usable portion of lakes, rivers, soil moisture and shallow groundwater: about 200,000 km³, under 1% of all fresh water.

Figure 1. Distribution of Earth's water. Data: Shiklomanov (1993) in Gleick (ed.), *Water in Crisis*; usable-portion estimate from UNEP (2002).

Fresh water is a flow renewed every year by the water cycle, not a fixed tank. About 47,000 km³ a year runs off the land or recharges groundwater (Gleick 1993, via UNEP 2002), of which about 12,500 km³ is geographically and seasonally accessible (Postel, Daily and Ehrlich 1996). Global withdrawals reached just under 4,000 km³ in 2021, up 14% from 3,500 km³ in 2000 (UNESCO/UN-Water 2025). Humanity therefore takes roughly a third of the accessible renewable supply. The share is indicative: it combines the 2021 FAO withdrawal figure with the runoff estimate of Postel et al., who themselves estimated withdrawals of about 4,430 km³ against the same accessible supply.

There is no single global date when fresh water runs out. What does run out are stocks that behave like tanks rather than flows: groundwater withdrawn faster than it recharges, glaciers losing mass, and rivers fully allocated in dry years (Section 3.5). The global average also conceals the defining fact of water scarcity, which is that it is local. Twenty-five countries, home to a quarter of the world's population, already use more than 80% of their available supply every year (Kuzma et al. 2023).

3.2 Where human water goes

Agriculture dominates human withdrawals (Figure 2), and in lower-income countries 90% or more of withdrawals go to irrigation (UNESCO/UN-Water 2025). Within food production, livestock is the largest user. Across all farm animals, Mekonnen and Hoekstra (2010)

estimated a water footprint of about 2,422 km³ a year, of which 87.2% is green water (rain falling on pasture and feed crops), 6.2% blue water and 6.6% grey water. Only the blue component, about 150 km³ a year, is consumptive use of surface and groundwater, the measure directly comparable with data-centre consumption; grey water is a notional volume needed to dilute pollution. Beef is the most water-intensive common food, with a global average footprint of about 15,400 litres per kilogram against 6,000 for pork and 4,300 for chicken, about 94% of it green water (Mekonnen and Hoekstra 2012), and beef cattle account for about one-third of the livestock total. These estimates cover 1996–2005; world beef and buffalo meat production has since risen to about 77 million tonnes (FAOSTAT 2024), so current volumes are likely higher.



Figure 2. Global freshwater withdrawals by sector, 2021. Data: UN World Water Development Report 2025, from FAO AQUASTAT.

A further 126 km³ of treated water, an average of 29.5% of supply, is lost every year from water distribution networks before reaching any user, mainly through leaks. Losses range from 4% of supply in Singapore to 83% in Armenia (Liemberger and Wyatt 2019). Where these losses are compared with data-centre withdrawal, the two differ in kind: about 90% of the IEA's 5.5 km³ data-centre withdrawal is returned to its source rather than consumed (0.56 km³ is consumed), whereas network losses are treated water that never reaches a user. Table 4 converts these volumes into average daily use per person.

Item	Litres per person per day	Basis
All human withdrawals	~1,370	4,000 km ³ ÷ 8.0 billion ÷ 365
Domestic and municipal	~180	13% of total
Livestock production, blue water	~51	150 km ³ ÷ 8.0 billion ÷ 365
Lost from supply networks	~43	126 km ³ ÷ 8.0 billion ÷ 365
All data centres, withdrawal, 2023	~1.9	5.5 km ³ ÷ 8.0 billion ÷ 365
All data centres, consumption, 2023	~0.19	0.56 km ³ ÷ 8.0 billion ÷ 365

Table 4. World average daily water use per person, derived. A single population of 8.0 billion (UN World Population Prospects 2022) is used for every row.

3.3 AI in proportion

Published estimates of data-centre and AI water demand are summarised in Table 5.

Estimate	Value	Measure	Source
Global data centres, 2023	5.5 km ³	Withdrawal	IEA 2025
Global data centres, 2023	0.56 km ³	Consumption	IEA 2025
Global data centres, 2030 projection	1.2 km ³	Consumption	IEA 2025
Direct cooling only, 2025 value and 2030 projection	0.22 (2025) and 0.64 (2030) km ³	Consumption, direct	Rystad Energy 2026
US hyperscale data centres (472 facilities), annual	~0.3 km ³ , about three-quarters electricity-related	Consumption, operational	Guidi and Dominici 2026
Global AI demand, 2027 projection	4.2–6.6 km ³	Withdrawal	Li et al. 2025
AI systems only, 2025 estimate	0.31–0.76 km ³	Footprint	de Vries-Gao 2025
AI data centres, 2030 projection	9.3 km ³	Footprint	Aczel et al. 2026
US data centres, 2023: direct vs indirect	Indirect ~12× direct	Consumption	LBNL, as reported by Gaster 2026

Table 5. Estimates of data-centre and AI water demand. The estimates measure different things and should not be added together. About two-thirds of the IEA's 2023 estimate (373 of 560 billion litres) is indirect, through electricity generation, and about a quarter (140 billion litres) is consumed on site (de Vries-Gao 2025); the US ratio is higher because it covers a different grid mix and boundary.

Figure 3 sets these estimates against the other water volumes described above. The scale is logarithmic: each gridline is ten times the previous one, which is the only way the data-centre values remain visible beside the others.

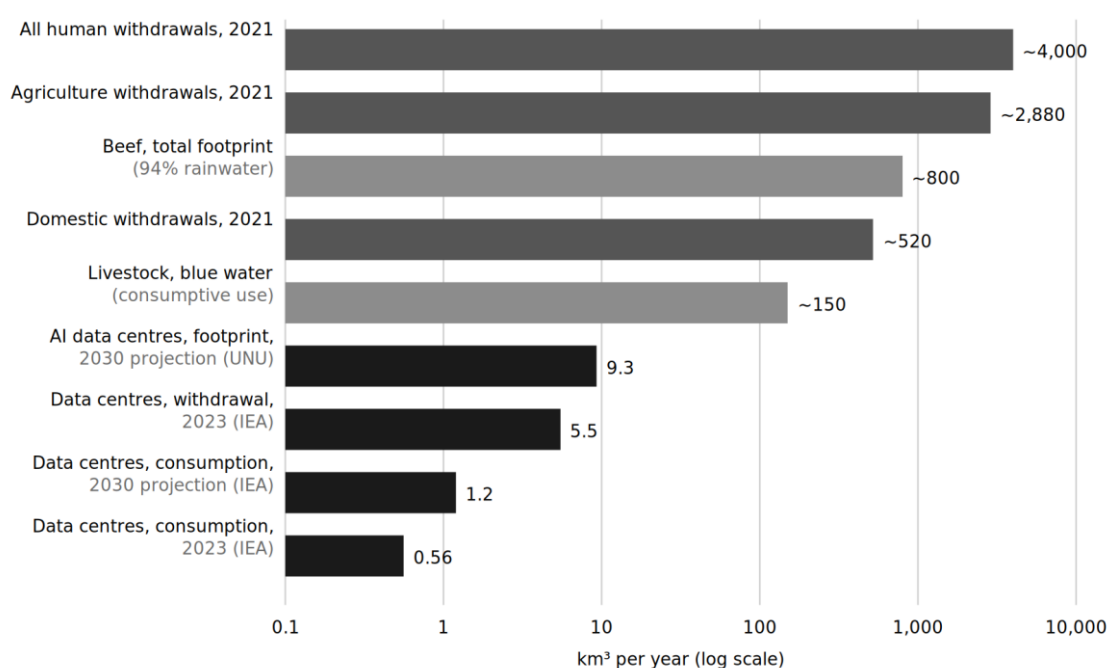


Figure 3. Annual water volumes compared. Dark bars are AI and data centres. Measures differ between rows (labelled); the chart shows orders of magnitude, not like-for-like accounting. Sources: WWDR 2025; Mekonnen and Hoekstra (2010, 2012); Aczel et al. (2026); IEA (2025).

Even the highest projection, the UNU-INWEH footprint for 2030, remains below a quarter of one percent of current human withdrawals. On global numbers alone, there is no

foreseeable point at which AI and people compete for the planet's water. Within individual companies the trend is still steep: Google's water consumption rose about 20% from 2021 to 2022 and a further 17% in 2023, when its self-operated data centres withdrew about 29 billion litres, nearly 80% of it potable (company sustainability reports, as compiled by Talukder et al. 2026). The global conclusion changes completely once location is added.

3.4 Siting

Water is used where it is found, and its scarcity is local. A 100 MW data centre in the United States consumes about 2 million litres a day, equal to about 6,500 households (IEA 2025, as reported by Nicoletti et al. 2025). Placed in a wet basin, that is marginal. Placed in a basin already using 80% or more of its supply, it competes directly with drinking water and farming. The evidence on where data centres are being built is summarised in Table 6.

Finding	Source
One-fifth of US data-centre servers draw their direct water from moderately to highly stressed watersheds; nearly half are powered fully or partly by plants in water-stressed regions	Siddik et al. 2021
Fewer than one-third of data-centre operators measured their water consumption; in some cases 57% of cooling water was potable	Mytton 2021
About two-thirds of new US data centres built or in development since 2022 are in places with high water stress; more than 160 new AI data centres in high-stress US areas in three years, up 70%	Nicoletti et al. 2025 (journalism; data to Q1 2025)
52% of global data-centre hubs are in high water-stress regions, projected to reach 58% by 2050	Baringa 2025 (industry analysis using WRI Aqueduct)
All data centres in Iran, Israel, Saudi Arabia and the UAE are in extremely high water-stress areas	Planet Tracker (not peer-reviewed)
Water-related protests against data centres in the Netherlands, Uruguay and Chile	Nicoletti et al. 2025

Table 6. Evidence on data-centre siting and water stress.

Siting follows cheap land, cheap electricity, tax incentives, fibre connections and proximity to users. Water rarely enters the decision because it is rarely priced at its scarcity value. In southern Nevada, the regional water authority attributed the spread of evaporative cooling to the price of water having been lower than the price of power (Las Vegas Sun 2023). Siddik et al. (2021) showed that locating US data centres differently could reduce their combined water and carbon footprints: the problem is a choice, not a technical necessity.

Local burden can be measured directly against the capacity of the utility that serves a facility. Across ten US data-centre sites, Akinade et al. (2026) found that burden ranged from 0.2% to 134% of host capacity: negligible for some communities, larger than the entire system for others. Han et al. (2026) estimate that, if 2024 water-use intensity persists, US data centres could require hundreds of millions of gallons a day of new public water-system capacity by 2030, with demand highest on the hottest days, when public systems may already be under strain. The two pathways also fall in different places: cooling burdens concentrate in stressed western and south-central basins, electricity-related burdens in a few grid regions (Guidi and Dominici 2026). A national share of 0.14%

can therefore coexist with local overload. These local-burden estimates are recent preprints; the one peer-reviewed siting study in this evidence base (Siddik et al. 2021) points in the same direction.

3.5 The systems being drawn on

Table 7 summarises the condition of the water systems into which this demand is being added.

Indicator	Value	Source
Countries facing extremely high water stress every year	25 (one-quarter of world population)	Kuzma et al. 2023
People under high stress at least one month per year	~4 billion	Kuzma et al. 2023
MENA population under extremely high stress	83% now; 100% by 2050	Kuzma et al. 2023
Additional people under extremely high stress by 2050	+1 billion	Kuzma et al. 2023
Aquifer systems declining >10 cm per year	36%	Jasechko et al. 2024
Aquifer systems declining >50 cm per year	12%	Jasechko et al. 2024
Regional aquifers with accelerating decline over four decades	30%	Jasechko et al. 2024
Glacier mass lost by 2100 (vs 2015), +1.5 °C / +4 °C	26 ± 6% / 41 ± 11%	Rounce et al. 2023
Glaciers disappearing by number, +1.5 °C / +4 °C	49 ± 9% / 83 ± 7%	Rounce et al. 2023

Table 7. Water stress, groundwater and glacier indicators.

Groundwater supplies roughly half of the water used for drinking and irrigation worldwide. The steepest declines are under cropland in dry climates, and median declines exceed a metre per year in regional aquifers under Iran, India, Spain, the United States, Saudi Arabia, Morocco and Mexico (UN-IGRAC summary of Jasechko et al. 2024). Glaciers are a critical water resource for about 1.9 billion people (Rounce et al. 2023); as they shrink, runoff first rises and then falls permanently. Part of their storage function can be replaced by reservoirs in basins becoming ice-free, up to $355 \pm 105 \text{ km}^3$ at suitable sites (Farinotti et al. 2019).

3.6 Case study: Egypt

Egypt combines the three conditions examined in this paper: a water supply with almost no internal source, an upstream situation without a binding operating agreement, and active recruitment of data-centre investment. It also has, unusually, the means to decouple data-centre growth from the Nile: a national desalination programme, long coastlines and proven large-scale solar generation.

3.6.1 A single-source water system

98% of Egypt's renewable water originates outside its borders (Egyptian

Ministry of Water Resources and Irrigation, as reported by *The National*, 21 September 2026). About 86% of the Nile's flow comes from the Ethiopian plateau, through the Blue Nile, Sobat and Atbara (Fanack Water 2026). Egypt's share under the 1959 agreement with Sudan is 55.5 billion m³ a year, a figure that has not changed since. Table 8 sets out the resulting water balance.

Indicator	Value	Source
Total actual renewable water resources; dependency ratio	58.3 billion m ³ ; 97%	FAO n.d.-a
Annual renewable water supply	~56.8 billion m ³	Ministry of Water Resources and Irrigation, Sep 2026
Estimated annual water needs	~123 billion m ³	Ministry of Water Resources and Irrigation, Sep 2026
Gap between renewable supply and needs	~66 billion m ³	Derived
Nile allocation, 1959 agreement	55.5 billion m ³	1959 Nile Waters Agreement
Share of renewable water from outside the borders	>98%	Ministry of Water Resources and Irrigation, Sep 2026
Per-capita availability	~472 m ³ /yr	Ministry, Sep 2026; absolute scarcity threshold 500 m ³

Table 8. Egypt's water balance.

Availability per person has fallen more than fivefold since 1947, while Egypt's Nile allocation has remained fixed at 55.5 billion m³ since 1959 (Figure 4). It is now below the threshold of absolute scarcity.

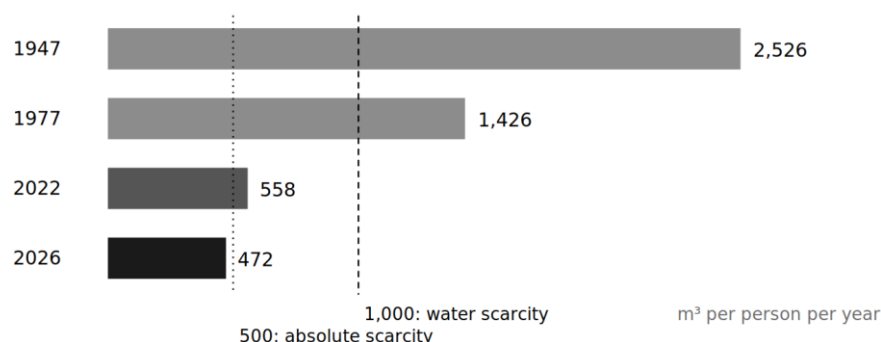


Figure 4. Renewable water per person in Egypt. Sources: CAPMAS (1947), via climate-diplomacy.org; MEED (2024), for 1977 and 2022; Ministry of Water Resources and Irrigation (2026). Sources use differing methods; the trend, not the exact values, is the point of comparison.

3.6.2 Upstream developments

Ethiopia inaugurated the Grand Ethiopian Renaissance Dam (GERD) on the Blue Nile on 9 September 2025. Its reservoir holds 74 billion m³, of which 59.2 billion is active storage, and its average annual outflow is estimated at about 50 billion m³ (Webuild n.d.). Figure 5 sets that storage against Egypt's renewable supply, its estimated needs and its allocation under the 1959 agreement.

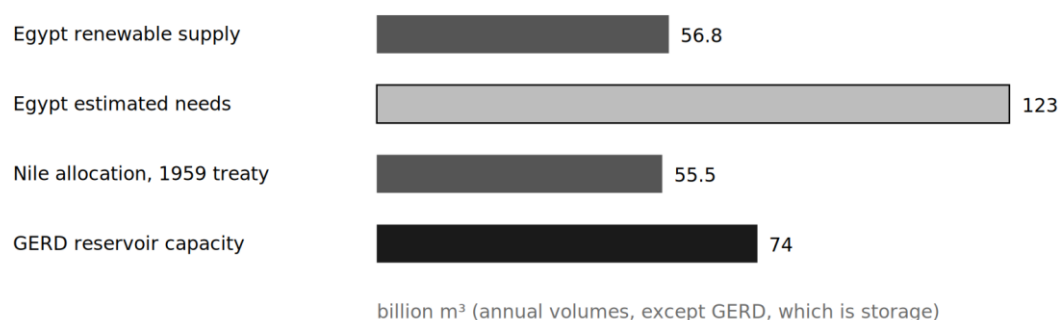


Figure 5. Egypt's water balance and upstream storage. Sources: Egyptian Ministry of Water Resources and Irrigation via *The National* (21 Sep 2026); 1959 Nile Waters Agreement via Al Jazeera; GERD specifications via Webuild and NPR.

In March 2026 Ethiopia announced three further Blue Nile dams, Karadobi, Mandaya and Beko Abo, with about 5,700 MW of combined capacity, still at planning stage (*The New Arab* 2026; Al Jazeera 2026). If all are built, combined upstream storage would be about three times Egypt's annual allocation (Table 9).

Project	Storage	Status	Source
GERD	74 billion m³	Operating since Sep 2025; 5,150 MW	NPR; Webuild
Karadobi, Mandaya, Beko Abo	30–40 billion m³ each (estimated)	Announced Mar 2026; planning stage	Egyptian experts via <i>The National</i> ; Al Jazeera
Combined upstream storage if all are built	~164–194 billion m³	About three times Egypt's annual Nile allocation	Derived from the above

Table 9. Blue Nile storage. The storage estimates for the three planned dams are unconfirmed.

Both positions are real. Ethiopia builds these dams for electricity in a country where almost half the population lacks power (NPR 2025), and a hydropower dam passes stored water downstream through its turbines, although reservoirs also lose water to evaporation; FAO (n.d.-b) estimates evaporation and other losses from Egypt's own High Dam lake at about 10 km³ a year. Egypt's objection concerns control: there is no binding agreement on filling and on releases during multi-year drought, and Egypt has taken its protest to the UN Security Council (Disruption Banking 2026). This paper takes no position on the dispute. The relevant point is structural: upstream storage capable of holding about three years of Egypt's allocation will operate without an agreed rule for sharing water under stress.

3.6.3 Data-centre growth

Egypt is actively recruiting data-centre investment. Table 10 lists the principal developments of 2026.

Date	Development	Source
June 2026	National Telecommunications Regulatory Authority licenses Hassan Allam Digital Infrastructure to build and operate data centres; initial investment about US\$400 million	Sada Elbalad, Aug 2026
September 2026	Government explores an integrated hyperscale data-centre and AI zone with Heka Data and a US consortium of developers	Business Today Egypt; Amwal Al Ghad, Sep 2026
September 2026	Minister reviews an 80 MW, US\$270 million first-phase data centre in the Sokhna industrial zone, Suez Canal Economic Zone (SCZONE)	TechAfrica News, 23 Sep 2026

Table 10. Recent data-centre developments in Egypt.

Scaled from the IEA's US average, an 80 MW facility cooled by evaporation would consume about 1,600 m³ a day, and a 1 GW campus about 20,000 m³ a day (Table 2). Because the water use of evaporative cooling varies strongly with climate and season (Karimi et al. 2022; Li et al. 2025), and Egyptian sites are hotter than the US average, these scaled volumes are more likely to understate than to overstate Egyptian demand. At national scale they are nonetheless negligible: roughly 0.001% and 0.013% of Egypt's renewable supply. The case for concern does not rest on national shares. A 1 GW campus would require about 1.5% of Egypt's entire current desalination capacity, and no public data allow its burden to be measured against the utilities that would serve it; US evidence shows that such burdens range from negligible to larger than the host system (Section 3.4). The risk for Egypt lies in the cumulative demand of a hyperscale zone and in precedent: whether the first facilities are licensed on Nile-derived supply will set the model for those that follow, for two decades.

3.6.4 Existing capacity for an alternative

Egypt already has much of the capacity that a non-potable supply for data centres would require (Table 11).

Capacity	Value	Source
Current desalination capacity	~1.3 million m ³ /day	Newsbase; national tender programme, 2026 (MEED reported about 0.8 million in 2024)
National desalination target, 2050	~8.8 million m ³ /day (~3.2 billion m ³ /yr)	Ministry of Housing, via Zawya; Smart Water Magazine
Planned investment in the desalination programme	~US\$8 billion	Deputy Minister of Housing, via Zawya (2023)
Desalination and wastewater PPP tenders	~US\$1 billion, including SCZONE projects, 2026	Newsbase; TendersGo
Programme priorities	Reverse osmosis, renewable energy, AI-based plant management, local membrane manufacturing	Sada Elbalad via TV BRICS, Jul 2026
Benban Solar Park	~1.8 GW; more than 4 TWh/yr	NREA data, as reported by Ahram Online and NS Energy
80 MW data centre cooling need as share of current desalination capacity	~0.12%	Derived: 1,600 ÷ 1,300,000 m ³ /day

Table 11. Egypt's desalination and solar capacity.

Egypt's national desalination strategy already prioritises renewable power and AI-based plant management, and its 2026 tender programme includes projects in the same economic zone where data centres are being planned. The cooling need of an 80 MW facility would equal about a tenth of one percent of the country's current desalination capacity. The infrastructure, the policy direction and the energy resource needed to keep data centres off the Nile already exist. What is missing is the requirement that links them.

4. Discussion

4.1 A systems reading

Governance of a common-pool resource. Hardin (1968) described how users of an open-access resource, each acting rationally, can collectively exhaust it. Most water used by data centres is not open-access: in Egypt the Nile allocation is set by treaty and distributed by the state, and in the United States water is supplied by utilities under public regulation. The more relevant framework is Ostrom's (1990): common-pool resources can be managed sustainably where rules, monitoring, accountability and prices reflect scarcity, and are overused where they do not. The evidence in Section 3 points to weakness on each condition. Water is priced below its scarcity value, as the southern Nevada case shows; fewer than a third of operators measured their water use at the time of Mytton's (2021) study; and UNU-INWEH now calls for standardised reporting because the data do not yet exist (Aczel et al. 2026). An operator choosing a site where water is cheap and unmeasured acts rationally within these rules. The rules are the problem.

Siting lock-in. Lock-in describes how long-lived infrastructure, the institutions around it and sunk investment reinforce one another until change becomes costly (Unruh 2000); large capital costs and long infrastructure lifetimes make it especially likely (Seto et al. 2016). For data centres it operates at two levels. Cooling systems can be retrofitted within a facility's life, and Track A relies on this. Location cannot: a facility built inland in a stressed basin cannot later switch to seawater, and one built far from a wastewater treatment plant cannot easily switch to recycled water. With design lives of 15 to 20 years (industry design estimates), a site chosen in a stressed basin today fixes the set of available water sources into the mid-2040s, when projections place the entire MENA population under extremely high water stress.

Single points of failure and shifted burdens. A resilient system has redundancy. Egypt's has little: more than 98% of its renewable water arrives through one river, and adding a new, fast-growing and long-lived demand removes whatever margin remained. Technical fixes can shift the burden rather than remove it. In a study of two data centres in Phoenix, the facility using air-cooled chillers had a power usage effectiveness about 13% worse than the evaporatively cooled one, but a source water usage 66% lower (Karimi et al. 2022). Saving water on site costs electricity, and electricity costs water elsewhere. UNU-INWEH found more broadly that choices which look best for carbon often turn out worse for water or land (Aczel et al. 2026).

Precedent. Humans have exceeded local water limits before: irrigation-driven salinisation in southern Mesopotamia has been linked to agricultural decline between about 2400 and 1700 BCE (Jacobsen and Adams 1958), although the extent of its role remains debated

(Powell 1985). What differs today is the scale and speed at which infrastructure fixes such choices across whole regions.

4.2 Trajectory

Reported incidents of water-related violence rose from 24 in 2000 to 420 in 2024, a 78% increase since 2022 alone, with the Middle East, South Asia and Sub-Saharan Africa most affected (Pacific Institute 2025). Most are local disputes over access, not wars between states. Between nations, cooperation still dominates: since 1948, 77% of recorded interactions over shared waters have been cooperative and 19% conflictual (Transboundary Freshwater Diplomacy Database, via New Security Beat 2023). What separates the two outcomes is institutional capacity. Wolf, Yoffe and Giordano (2003) found that most parameters commonly cited as indicators of water conflict are only weakly linked to disputes, while institutional capacity, in the form of treaties, management bodies and positive international relations, matters as much as physical conditions or more. The Basins at Risk project summarises the mechanism: the likelihood of conflict rises when the rate of change in a basin exceeds the institutional capacity to absorb it, and a major project developed without a treaty or joint commission is among its clearest indicators. Rapid growth of water-dependent infrastructure in basins without agreed rules fits that description. This paper does not claim that data centres cause conflict. The point is narrower: the condition associated with rising risk, change that outpaces institutions, is the same condition under which data-centre capacity is now being added in several stressed basins. That gives the institutional measures in Section 5 a relevance beyond water efficiency.

4.3 The window

The timelines of the stressed systems now converge with the lifetime of infrastructure being built today (Table 12).

Timeline	What happens
2025 to 2030	Data-centre water consumption projected to roughly double (IEA) or nearly triple for direct cooling (Rystad); site decisions for this capacity are being made now
Mid-2040s	Facilities sited today reach the end of a 15 to 20 year design life
By 2050	100% of the MENA population under extremely high water stress; +1 billion people globally; three-quarters of new glacier-basin storage potential ice-free
By 2100	26% to 41% of glacier mass lost depending on warming

Table 12. Converging timelines.

Problems of this kind are easiest to solve while a system still has slack. Once scarcity arrives, attention, money and political capacity move to managing the emergency: rationing, emergency supply, disputes over allocation. The long-term fixes that would have prevented the crisis then compete with the crisis itself, and usually lose. The Nile shows the pattern: a question that could have been settled by technical agreement while the dam was being planned is now argued at the UN Security Council, in the language of security rather than cooperation. The window in which siting can be changed cheaply is the current construction cycle.

5. Recommendations

Every element needed to decouple AI infrastructure from scarce fresh water already exists and is operating somewhere. What is missing is the rule that makes it standard practice.

Three tracks follow. Each addresses a different failure identified in Section 4: Track A the competition between data centres and people for the same fresh water, Track B the lock-in created by siting decisions made on present rather than future conditions, and Track C the absence of any mechanism through which the AI sector contributes to the water systems it draws on. The tracks are designed to operate together, and each is set out in the same structure: objective, scope, implementation, responsibilities, cost, monitoring and risks.

Sequencing. The tracks presuppose institutions that, as Section 4.1 notes, are often weakest where data-centre growth is fastest. They are therefore ordered by the capacity they require. Measures applied once, at a single point, come first: the source requirement of Track A and the lifetime assessment of Track B can be attached as conditions to the licence or permit that every jurisdiction already issues. Annual reporting follows, and builds the data needed for the third stage: basin-level accounting and the verification on which Track C depends.

5.1 Track A: non-potable cooling water and low-water electricity in water-stressed basins

Objective. To remove direct competition between data-centre cooling and drinking or irrigation water in basins that have no margin left. Cooling water only needs to remove heat. It does not need to be drinkable, yet in some cases more than half of data-centre cooling water has been potable (Mytton 2021), and nearly 80% of Google's 2023 on-site withdrawals were potable (company reports, as compiled by Talukder et al. 2026).

Scope and phasing. Phase 1 applies to new facilities in basins classified by WRI Aqueduct as having extremely high baseline water stress (over 80% of available supply withdrawn), using the projection for the end of the facility's design life rather than current conditions. Phase 2 extends the standard to high-stress basins (40–80%) once Track B reporting has established how much capacity they contain; about half of global data-centre hubs are estimated to lie in high-stress regions (Baringa 2025), so the second phase should rest on that evidence. Existing facilities in basins covered by the current phase convert at their next major replacement of cooling plant or by a nationally set deadline in the near term, whichever comes first. Cooling plant typically lasts 15 to 30 years (industry sources), longer than many facilities' design lives, so replacement alone could leave some facilities unconverted for their entire operating life; the deadline closes that gap, while the replacement trigger avoids stranding working equipment where it comes sooner. The capacity threshold above which the standard applies is set nationally.

Two components. On-site water accounts for only about a quarter of data-centre water consumption; about two-thirds is consumed in generating the electricity (de Vries-Gao 2025), and at US hyperscale facilities roughly three-quarters of operational water is electricity-related (Guidi and Dominici 2026). A non-potable standard on its own therefore addresses the smaller share. Track A has two components: A1, a non-potable standard for

on-site cooling, and A2, electricity for facilities in mapped basins drawn increasingly from low-water sources. Photovoltaics and wind have the lowest operational water consumption of all generating technologies (Macknick et al. 2012), so A2 reinforces decarbonisation rather than competing with it.

Order of preferred sources (A1). The standard would set a hierarchy rather than a single technology (Table 13).

Priority	Source	Condition
1	Treated municipal or industrial wastewater	Where a treatment plant with sufficient output is within reach
2	Direct seawater cooling through heat exchangers	Where sea temperatures allow effective heat rejection
3	Dedicated or shared seawater desalination powered by renewable energy	Coastal sites where direct seawater cooling is ineffective
4	Air or dry cooling with no water use	Where grid capacity and low-carbon power allow the higher electricity demand
5	Potable water	Emergency backup only, with each use reported

Table 13. Source hierarchy under Track A.

Each non-potable route already operates at commercial scale (Table 14).

Route	Operating example	Limitation
Recycled wastewater	Google, Douglas County, Georgia: switched to a fully recycled supply, taking treated municipal wastewater before it returns to the Chattahoochee River and treating its own effluent before discharge	Requires a nearby treatment plant with enough output
Direct seawater cooling	Google, Hamina, Finland (since 2011): cooled with Gulf of Finland water through heat exchangers, without chillers; water is tempered before return	Depends on cold water; warm surface seas such as the Red Sea and the Gulf cool far less effectively; seawater corrodes equipment
Dedicated renewable-powered desalination	Seawater reverse osmosis is mature; 15,906 plants produce about 95 million m ³ a day, 48% of it in the Middle East and North Africa (Jones et al. 2019)	Energy use and brine (below)

Table 14. Non-potable cooling routes. Sources: DatacenterDynamics; Data Center Knowledge; Wired/Gizmodo reporting on Douglas County; Jones et al. 2019.

Where the sea is too warm for direct cooling, it can still be the source. The calculation in Table 15 shows that desalinating all on-site data-centre water would be a small cost at sector scale.

Item	Value	Basis
Energy to desalinate seawater by reverse osmosis	2.5–3.5 kWh/m ³	Typical current range (Ritchie 2024); record 1.79 kWh/m ³ in 2025 (Gran Canaria); theoretical minimum about 1 kWh/m ³
On-site data-centre withdrawal, 2030 projection	~0.38 km ³	Table 2
Energy to desalinate it	~1.3 TWh/yr	Table 2
Data-centre electricity, 2030 projection	~945 TWh	IEA 2025
Energy penalty	~0.14%	0.07–0.19% under alternative assumptions (Table 3)
Equivalent continuous power	~150 MW	1.3 TWh ÷ 8,760 h
Annual water cost	~US\$0.12–0.56 billion	From US\$0.31/m ³ (Hassyan, UAE tariff; TRENDS 2025) to US\$1.50/m ³ (Ritchie 2024)
Comparison: annual output of Egypt's Benban solar park (~1.8 GW, Aswan)	>4 TWh/yr	NREA data, as reported by NS Energy
Share of current global desalination output	~1.1%	0.38 of 34.7 km ³ /yr (95 million m ³ /day; Jones et al. 2019)

Table 15. Feasibility of supplying data centres' on-site water from seawater desalination.

Supplying all on-site data-centre water from the sea would add about 0.14% to the sector's projected electricity demand (Figure 6).

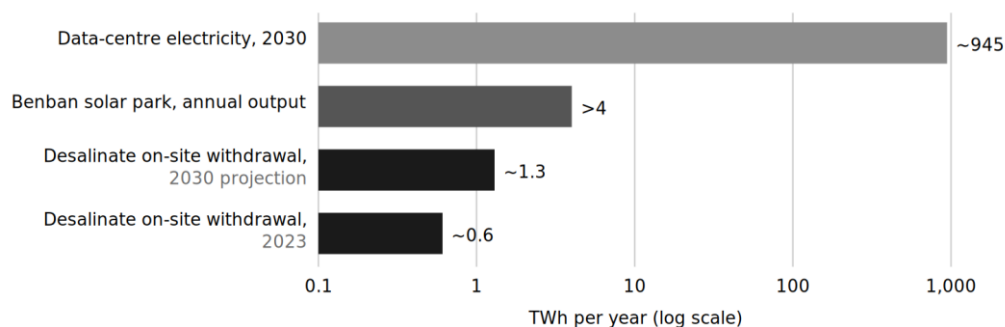


Figure 6. Energy needed to supply all data-centre on-site water by seawater desalination, compared with the output of one large solar park and the sector's projected electricity use. Log scale. Sources: IEA (2025); de Vries-Gao (2025); NS Energy; derived at 3.5 kWh/m³ (Table 2).

The energy needed is about a third of the annual output of a single existing solar park, Benban in Egypt. The barrier is not cost or physics. It is that fresh water in most locations is cheaper still, because its scarcity is not priced.

Energy supply. Coastal sites have sun and wind now and, in the longer term, wave and tidal energy (Table 16).

Energy source	Status	Role for coastal sites
Solar and wind	Mature, commercially deployed at scale	Available now
Tidal	Most of the world's ~513 MW ocean capacity is two tidal barrages (La Rance 240 MW; Sihwa 254 MW); tidal stream in pre-commercial farms	Predictable output; suitable sites limited
Wave	Early stage; two full-scale devices totalling 130 kW deployed in Europe in 2024; 1.6 MW of ocean capacity added worldwide that year	Large theoretical potential; costs projected to fall to about US\$120/MWh once 2 GW is deployed (IRENA)

Table 16. Ocean-adjacent renewable energy. Sources: REN21 2025; Ocean Energy Europe 2024; IRENA.

Implementation. The steps are set out in Table 17.

Step	Action
1	National water authorities map basins meeting the stress criterion over a 20-year horizon, using WRI Aqueduct projections or national equivalents, and publish the map
2	Permitting authorities attach the source hierarchy as a licence condition for new data centres in mapped basins
3	Where desalination is the chosen route, facilities connect to or co-fund shared municipal or industrial-zone desalination plants where these exist, rather than building isolated units, to gain economies of scale and consolidate brine management
4	Brine discharge is licensed separately, with limits on salinity and temperature at the outfall, diffuser requirements and seabed monitoring
5	Existing facilities in mapped basins submit a conversion plan, due at their next major cooling-plant replacement or by the national deadline, whichever comes first
6	Operators in mapped basins report the water intensity of their electricity supply and set a schedule for shifting it toward low-water sources (A2)

Table 17. Track A implementation steps.

Brine. Desalination produces about 1.5 litres of concentrated brine per litre of fresh water; global output is about 142 million m³ a day, 55% of it from Saudi Arabia, the UAE, Kuwait and Qatar (Jones et al. 2019). Brine is denser than seawater, settles on the seabed, carries less oxygen and can contain treatment chemicals. Supplying all on-site withdrawals at the projected 2030 level would add about 0.56 km³ of brine a year, about 1% of current global brine output. Track A is therefore conditional on step 4: without discharge standards, it would move the burden from fresh water to the sea.

Local conditions matter more than global totals. In the Gulf of Aqaba, reef-building corals were significantly impaired, with partial bleaching, by salinity only 10% above ambient combined with antiscalants (Petersen et al. 2018). Discharge into the Gulf of Suez and the Red Sea therefore requires hydrodynamic modelling, outfalls sited away from reef habitat and long-term monitoring.

Responsibilities. National water authorities (mapping and source hierarchy); permitting and investment authorities, including special economic zones (licence conditions); environmental and marine authorities (brine licensing); operators (compliance and reporting).

Cost. About 0.14% of projected sector electricity (0.07% to 0.19% under alternative assumptions) and about US\$0.12 to 0.56 billion a year in water cost at global scale if all on-site withdrawals were desalinated (Tables 2 and 3). Where recycled water or direct seawater cooling is available, the cost can be lower still: Google's Hamina and Douglas County facilities adopted these routes in part for the operating savings they were reported to bring. A2 is implemented through the procurement of electricity supply.

Monitoring indicators. Share of cooling water from non-potable sources, by facility and basin; litres of potable water per kWh of IT load; share of electricity from low-water sources; brine volume and outfall salinity.

Risks. Brine impacts if step 4 is weak; higher electricity demand where air cooling is chosen, which shifts water use to power plants unless that power is renewable (Karimi et al. 2022); corrosion and maintenance costs for seawater systems; and relocation of facilities to unregulated basins, which Track B addresses.

5.2 Track B: siting over the full operating life, with complete reporting

Objective. To prevent lock-in by making the water conditions a facility will face over its whole life, not only today, part of the decision to build it, and to create the data without which neither regulators nor researchers can manage the problem.

Scope. All new data centres above the national capacity threshold, and annual reporting by all existing facilities above it.

Components. Track B has five components (Table 18).

Component	Content
Lifetime water-source assessment	Before a permit is granted: the water source, its projected stress over the design life, its reliability in drought years, and the share of the basin's remaining margin the facility would take
Indirect water	The assessment includes water consumed in generating the facility's electricity, based on the expected grid or contracted supply mix
Annual public reporting	On-site withdrawal and consumption, indirect water, and water use efficiency at both site and source level, broken down by source type (potable, groundwater, recycled, seawater, desalinated)
Basin-level accounting	Basin authorities track the combined demand of all data centres and other large new users against the basin's projected supply, so that cumulative effects are visible
Integration into national planning	AI infrastructure is included in national energy, water and land-use plans (Aczel et al. 2026)

Table 18. Track B components.

Reporting by source type matters because the most widely used efficiency metric, water usage effectiveness, treats a litre of recycled water and a litre of drinking water as equivalent, and operators report it in different ways; one operator, for example, publishes separate figures for its whole portfolio (0.95 litres per kWh in 2024) and for its evaporatively cooled sites only (1.55) (Equinix 2026). Without source-level reporting, Track A cannot be monitored. Existing precedents for Track B measures are listed in Table 19.

Measure	Precedent or evidence
Assess basin water stress over a facility's full design life, using projections rather than present conditions, before permits are granted	Siddik et al. 2021: relocation reduces combined water and carbon footprints; WRI Aqueduct provides 2030–2050 projections
Mandatory reporting of on-site and indirect water, by source (potable, recycled, seawater), alongside energy	Recommended by UNU-INWEH (Aczel et al. 2026); fewer than a third of operators measured water in 2021 (Mytton 2021)
Water-efficiency targets	Singapore Green Data Centre Roadmap: under 2 litres per kWh; EU Data Centre Energy Efficiency package with labelling in preparation (Rystad Energy 2026)
Restrictions on evaporative cooling in the most stressed basins	The Southern Nevada Water Authority banned evaporative cooling in new commercial and industrial buildings from September 2023 (Las Vegas Sun 2023; Nevada Independent 2026)
Integrate AI infrastructure into national energy, water and land-use planning	Recommended by UNU-INWEH (Aczel et al. 2026)

Table 19. Track B measures and precedents.

Implementation. The steps are set out in Table 20.

Step	Action
1	A standard reporting template covering the elements in Table 18 is adopted, ideally aligned internationally so that multinational operators report once
2	Permitting authorities require the lifetime water-source assessment as part of the application for new facilities
3	Operators of existing facilities begin annual reporting within a set period
4	Basin authorities publish combined data-centre demand alongside other large users
5	Reported data are made publicly available to allow independent research

Table 20. Track B implementation steps.

Responsibilities. Permitting and investment authorities (assessment requirement); water and basin authorities (basin accounting); statistical or environmental agencies (reporting standard and publication); operators (data).

Cost. Administrative. The main cost is the assessment itself, small relative to investments of hundreds of millions of US dollars per facility.

Monitoring indicators. Share of new capacity with a completed lifetime assessment; share of facilities reporting; combined data-centre demand as a share of each basin's projected supply.

Risks. Inconsistent reporting across jurisdictions, which step 1 addresses; commercial confidentiality claims, which can be met by reporting at basin rather than facility level where necessary; and assessments treated as formalities, which public disclosure makes harder.

5.3 Track C: AI applied to losses in the water systems it depends on

Objective. To give the AI sector a direct, measurable contribution to water security in the basins where it operates, directed at the largest avoidable loss in urban water supply.

The world's supply networks lose about 126 km³ of treated water a year, an average of 29.5% of supply, ranging from 4% in Singapore to 83% in Armenia (Liemberger and Wyatt

2019). That is about 23 times what all data centres withdraw (Figure 7). In North Africa and the Middle East alone, losses are nearly three times the global data-centre total. Reducing global losses by 115 million m³ a day has been estimated to require about US\$69 billion, with an annual benefit of about US\$13 billion and a payback period of 7 to 10 years (Mosarat et al. 2026). Leak detection combining sensor networks with machine learning is an active field of research and deployment, as is machine-learning optimisation of desalination plants, which Egypt's national desalination strategy already includes (Section 3.6.4).

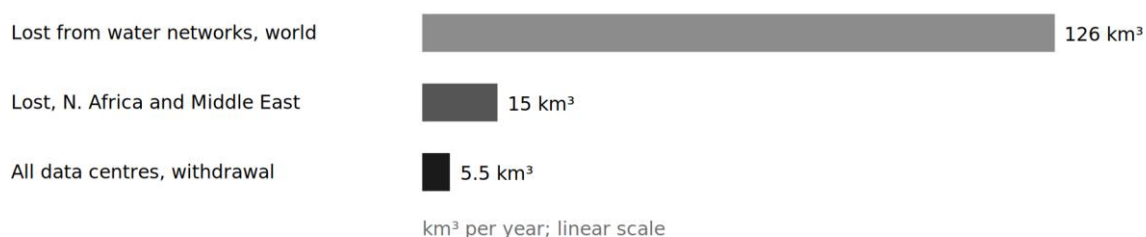


Figure 7. Treated water lost from supply networks compared with total data-centre withdrawals. Sources: Liemberger and Wyatt (2019); regional figure as tabulated in Mosarat et al. (2026); IEA (2025).

Mechanism. Operators in mapped basins would fund, or deliver in kind through AI tools and expertise, reductions in the local utility's network losses at least equal to a reference volume that follows the facility's stage under Track A. Before conversion, the reference is the facility's annual on-site freshwater withdrawal: withdrawal rather than consumption is used because it is the volume the utility must supply and the larger of the two, so the ratio errs on the side of the basin. After conversion, when on-site freshwater use falls to near zero, the reference becomes the water consumed in generating the facility's electricity within the same basin, as reported under Track B. The obligation therefore lasts as long as the facility draws on the basin's water, directly or indirectly. Only reductions additional to the utility's existing programme count: savings are measured against a baseline trajectory that includes work already planned or funded. Major operators have already set targets to become "water positive" by 2030, replenishing more water than they consume (Rystad Energy 2026). Track C makes such commitments local, additional and verifiable.

Implementation. The steps are set out in Table 21.

Step	Action
1	The utility serving each mapped basin establishes a baseline trajectory of network losses, including reduction work already planned or funded, using the International Water Association water-balance method
2	Operators in the basin agree with the utility a programme of loss reduction, such as sensors, AI-based leak detection and pressure management, sized to at least their reference volume (on-site freshwater withdrawal before conversion; in-basin electricity-related water after)
3	Where desalination plants serve the basin, operators may count verified energy or water savings from AI-based plant optimisation toward the same total
4	Savings are measured by the utility against the baseline and reported annually
5	Tools and data developed under the programme are made available to other utilities in the region

Table 21. Track C implementation steps.

Responsibilities. Water utilities (baselines, delivery and measurement); operators (funding, tools and expertise); regulators (verification).

Cost. At the scale of individual facilities, modest: an 80 MW evaporatively cooled facility withdrawing about 2,000 m³ a day (1,600 m³ consumed at 80% evaporation) would need to recover at least that volume from the utility's losses. Because reduced losses also reduce utilities' production and energy costs, part of the investment is recovered.

Monitoring indicators. Verified additional water recovered per operator per year, compared with that operator's reference volume; change in the basin's loss rate.

Risks. Double counting between operators, which a single utility-held register prevents; savings that are not sustained, which annual measurement addresses; and use of the mechanism as a substitute for Track A rather than an addition to it. Track C is not an offset licence: it applies on top of the non-potable standard, not instead of it.

5.4 Supporting measures

Shared governance. Groundwater has recovered where rules were applied: pumping regulation in Bangkok, surface-water transfer in Albuquerque and managed aquifer recharge in Arizona (Jasechko et al. 2024). The same principles, of clear rules, monitoring and accountability (Ostrom 1990), apply at basin scale to data centres and, across borders, to rivers such as the Nile, where the absence of a binding operating agreement is the core risk.

Beyond AI. Agriculture takes 72% of withdrawals. Irrigation efficiency, the water intensity of diets and network repair will do more for global water security than any change in the AI sector. This paper focuses on AI because its infrastructure is being sited now, and those decisions are still open.

5.5 Application to Egypt

The three tracks are applied to the Egyptian case of Section 3.6 in Table 22.

Track	Application	Existing basis
A	Data centres in SCZONE and other coastal zones cool with seawater-sourced water, connecting to or co-funding the zone's planned desalination plants, with brine outfalls sited away from reef habitat; electricity drawn increasingly from solar and wind	National desalination programme; SCZONE desalination tenders; Benban-scale solar generation
B	Data-centre licences, including the 2026 licensing round and the proposed hyperscale zone, require a lifetime water-source assessment and annual public water reporting	Existing licensing through the telecommunications regulator and SCZONE
C	Operators fund AI-based loss reduction in the networks serving their zone, with savings verified by the utility	AI-based plant management already in the national desalination strategy

Table 22. The three tracks applied to Egypt.

Egypt's position, with almost no internal water, long coastlines, proven large-scale solar generation, an established desalination programme and active data-centre recruitment,

makes it a place where these rules could be set before the first large facilities are built rather than retrofitted afterwards.

5.6 Summary of recommendations

Table 23 summarises the three tracks, the failures they address and their readiness for implementation.

Track	Addresses	Readiness
A. Non-potable cooling water (recycled, seawater, renewable desalination with brine limits) and low-water electricity	Competition with drinking and irrigation supply; indirect footprint	Operating examples; ~0.14% energy penalty for desalination; phased by stress level
B. Lifetime siting assessment and complete reporting	Lock-in in stressed basins; missing data	Policy precedents in Singapore, EU, Nevada
C. AI for water-network losses and plant optimisation, additional and at least equal to the water each facility draws from the basin	126 km ³ /yr of treated water lost	Active research and deployment

Table 23. Summary of recommendations.

6. Limitations

The evidence base was assembled through structured web, publisher and preprint searches rather than a systematic review of bibliographic databases, so relevant studies may have been missed. Some comparisons set different measures side by side; where they do, this is stated, and the headline comparisons use like-for-like measures (withdrawal against withdrawal, consumption against consumption). Global per-person values are averages that hide large regional variation. IEA water figures are cited as reproduced in de Vries-Gao (2025) and Nicoletti et al. (2025). The livestock water footprint relies on 1996–2005 data, and green-water accounting remains contested. The 2030 on-site share of data-centre water is assumed equal to the IEA's 2023 share; Table 3 tests a higher value. Evidence on siting patterns draws partly on journalism and industry analysis (Bloomberg, Planet Tracker, Baringa), identified as such in the text, and the local-burden estimates are recent preprints. Desalination energy and cost vary with site, salinity, intake and brine handling. Direct seawater cooling does not transfer to warm seas. The Egyptian facility estimates are scaled from a US average, and no public data allow burden to be measured against the utilities that would serve Egyptian sites. Egyptian per-capita water figures come from sources using different methods, and the storage of the three planned Blue Nile dams is an unconfirmed estimate. The implementation designs in Section 5, including the phasing of Track A and the additionality rule in Track C, are proposals and have not been tested as a package in any jurisdiction. The conflict data count reported incidents, and part of the recent increase may reflect better reporting. The specific wording of the Basins at Risk mechanism is taken from the project's summaries; the core institutional-capacity finding is from Wolf, Yoffe and Giordano (2003). The Mesopotamian salinisation thesis remains debated. Projections to 2030 and beyond carry the uncertainties of their underlying models.

7. Conclusion

AI is not what is emptying the world's water supply. Agriculture takes 72% of withdrawals, livestock production consumes about 270 times as much surface and groundwater as every data centre combined, supply networks leak 23 times more than data centres withdraw, and aquifers were falling long before generative AI existed. Blaming AI misplaces responsibility and invites the wrong response.

The real problem is human: resources that are shared, underpriced and poorly measured, now receiving a new, fast-growing and long-lived demand in many of the places least able to absorb it, often in basins without the rules that govern competition for water.

The solutions are neither speculative nor prohibitively expensive. Data centres already run on recycled water and seawater. Supplying all on-site data-centre water from renewable-powered desalination would take less than 0.2% of the sector's projected electricity, and shifting its power supply toward solar and wind reduces the larger, indirect share. AI applied to leaking water networks could recover many times the water AI consumes. What is missing is the requirement to do these things, and the requirement is cheapest to set before the current generation of facilities is built. AI will remain part of how societies function. The task is to make sure it grows in a way that strengthens the water systems people depend on, so that there is enough for both.

Today there is still time, capital and relative stability to build these systems properly. Those conditions are themselves a resource, and like water they will not always be available. What is built with them now is what the future will have to work with.

Relevance to the Sustainable Development Goals

SDG 6 (clean water and sanitation), particularly Target 6.4 on water-use efficiency and reducing the number of people suffering from water scarcity, is directly affected by where water-intensive infrastructure is sited and by network losses.

SDG 9 (industry, innovation and infrastructure) calls for resilient infrastructure, which is incompatible with siting that removes a region's water reserve.

SDG 12 (responsible consumption and production) applies to the food-system comparison and to the AI sector.

SDG 13 (climate action) intersects through glacier loss and the carbon-water trade-offs of cooling.

SDG 14 (life below water) applies to the brine standards that desalination requires.

SDG 16 (peace, justice and strong institutions) applies through the link between institutional capacity and water-related conflict.

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Use of AI tools: An AI assistant (Claude, Anthropic) was used for data compilation, calculation checks, figure preparation and drafting support. The author directed the

research questions, the argument and the recommendations, and reviewed and takes full responsibility for all content.

Data availability: All data are from the publicly available sources cited. Derived values and their inputs are listed in Table 2.

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